

Einsendeaufgaben – Aufgabe 4.2

61211 Analysis

Aufgabe 1

a)

$$\begin{aligned}D_1 f(x, y) &= 2x \sin(x + y - 1) + (1 + x^2 + y^2) \cos(x + y - 1) \\D_2 f(x, y) &= 2y \sin(x + y - 1) + (1 + x^2 + y^2) \cos(x + y - 1) \\D_{11} f(x, y) &= 2 \sin(x + y - 1) + 2x \cos(x + y - 1) \\&\quad + 2x \cos(x + y - 1) - (1 + x^2 + y^2) \sin(x + y - 1) \\D_{12} f(x, y) &= D_{21} f(x, y) = 2x \cos(x + y - 1) + 2y \cos(x + y - 1) \\&\quad - (1 + x^2 + y^2) \sin(x + y - 1) \\D_{22} f(x, y) &= 2 \sin(x + y - 1) + 2y \cos(x + y - 1) \\&\quad + 2y \cos(x + y - 1) - (1 + x^2 + y^2) \sin(x + y - 1)\end{aligned}$$

$$\begin{aligned}f(a) &= \sin(-1) & D_{11} f(a) &= 2 \sin(-1) + \cos(-1) \\D_1 f(a) &= \cos(-1) & D_{12} f(a) &= D_{21} f(a) = -\sin(-1) \\D_2 f(a) &= \cos(-1) & D_{22} f(a) &= 2 \sin(-1) - \sin(-1)\end{aligned}$$

$$\begin{aligned}T_1(x) &= \sin(-1) + \cos(-1)x_1 + \cos(-1)x_2 \\T_2(x) &= T_1(x) + \frac{1}{2}(2 \sin(-1) + \cos(-1))x_1^2 \\&\quad - \sin(-1)x_1x_2 \\&\quad + \frac{1}{2}(2 \sin(-1) - \sin(-1))x_2^2\end{aligned}$$

b)

$$D_1g(x, y, z) = -\frac{y}{x^2} + yz - 2xz$$

$$D_2g(x, y, z) = \frac{1}{x} + xz$$

$$D_3g(x, y, z) = xy - x^2$$

$$D_{11}g(x, y, z) = \frac{2y}{x^3} - 2z$$

$$D_{12}g(x, y, z) = D_{21}g(x, y, z) = -\frac{1}{x^2} + z$$

$$D_{13}g(x, y, z) = D_{31}g(x, y, z) = y - 2x$$

$$D_{23}g(x, y, z) = D_{32}g(x, y, z) = x$$

$$D_{22}g(x, y, z) = 0$$

$$D_{33}g(x, y, z) = -2$$

$$T_1(x, y, z) = -1 - 2(x - 1) + 2y - (z - 1)$$

$$T_2(x, y, z) = T_1(x, y, z) - (x - 1)^2 - 2(x - 1)(z - 1) - (z - 1)^2$$